

Nonuniform external fields and vacuum properties in a two-dimensional gauge theory

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(Received 5 January 1982)

We compute the effective action for scalar electrodynamics in two-dimensional space-time, to the one-loop order, for a two-parameter solitonlike family of nonuniform electric fields. We find that the vacuum is always dia-electric for this class of fields. The electric permeability results to be a decreasing function of the field strength and an increasing function of the field extension.

I. INTRODUCTION

It is generally believed that quark confinement can occur owing to a nontrivial structure of the ground state. Recently, the effective action of non-Abelian gauge theories has been studied¹⁻⁴ in order to analyze the structure of the vacuum and decide if a nonzero external field may have a lower energy density than the perturbative ground state.

One of the problems in these investigations lies in the strong dependence of the results on the choice of the background field. Covariantly constant gauge fields have been shown to lower the energy of the QCD vacuum,¹⁻² but the resulting state becomes unstable.³ This vacuum can be approximately stabilized by separating the space into domains and allowing the fields to change from one domain to another,³ thus showing that spatial dependence can drastically change the results of effective-action calculations. The absence of vacuum polarization at the one-loop level, when non-Abelian planewaves are considered as a background field,⁵ also points in this direction.

It is then relevant to analyze the dependence of the effective action on external nonuniform fields in order to establish how much the resulting vacuum structure relies on the choice of the background field. In this domain one is faced, at the one-loop level, with the computation of functional determinants for spatially dependent fields, this leading naturally to the application of inverse scattering methods, a framework where calculations can be performed in a closed form.

It is well known that a one-dimensional functional determinant coincides with the Jost function of the associated scattering problem. Higher-dimensional determinants can still be expressed as integrals or infinite sums of Jost functions provided the external field depends on only one variable.^{6,7}

For the particular case of solitonlike fields (i.e., Bargmann-type potentials), determinants become simpler, the Jost functions being in this case rational fractions of Γ functions. It is precisely the aim of the present article to explore the vacuum of a gauge theory from the effective action for solitonlike external fields.

As a laboratory, we have chosen scalar electrodynamics in two space-time dimensions, a model which, being simpler than four-dimensional QCD, shares some of its relevant properties. In particular, a nontrivial vacuum structure has been found by studying the one-loop effective action for uniform electric field.⁸

In the present paper we shall consider nonuniform fields of the soliton type, depending on two parameters—one characterizing the field strength and the other its spatial extension. A general scheme based on Gelfand-Levitan equations can be envisaged to generate one-dimensional background fields, with associated solvable effective action, depending on an arbitrary number of parameters.

The paper is organized as follows. In Sec. II, we define the model and establish the relation between the one-loop contribution to the effective action and the Jost functions associated with a Klein-Gordon-type equation. We discuss the properties of the solutions and the associated scattering data and then, in Sec. III, we compute the one-loop effective action in a closed form. We introduce a dielectric permeability ϵ for nonuniform fields in terms of the effective Lagrangian. The results can be summarized first by saying that the vacuum exhibits dia-electricity ($\epsilon < 1$) in the presence of our nonuniform field. We find that ϵ decreases monotonically for large field strengths at fixed spatial extension. For fixed strength, ϵ grows monotonically (but remaining below one) when the spatial extension decreases. The renormalization-group

analysis shows that the field-strength dependence of ϵ is given by the Gell-Mann—Low β function as is the case for uniform fields for this model, as well as for QCD.¹ On the other hand, no new information about the dependence on the field extension follows from the renormalization-group analysis.

II. THE MODEL

We start from the two-dimensional (Euclidean) Lagrangian

$$\mathcal{L} = \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + (D_\nu \phi)^* D^\nu \phi + V(\phi), \quad (2.1)$$

where ϕ is a charged scalar field, $D_\nu = \partial_\nu - ieA_\nu$, $x_\nu = (x, y)$, and

$$V(\phi) = m^2 \phi^* \phi + \frac{1}{4} \lambda (\phi \phi^*)^2. \quad (2.2)$$

The generating functional for the one-particle irreducible Green's functions—the effective action—is given at the one-loop level by

$$\Gamma[A_\nu^{\text{cl}}(\cdot), \phi^{\text{cl}}(\cdot)] = \int d^2x \mathcal{L}(A_\nu^{\text{cl}}, \phi^{\text{cl}}) + \frac{1}{2} \int d^2x d^2x' \partial_\mu F_{\mu\alpha}^{\text{cl}}(x) M^{-1}_{\alpha\beta}(x, x') \partial_\nu F_{\nu\beta}^{\text{cl}}(x') + \ln \det M, \quad (2.3)$$

where M is the operator appearing in the quadratic term in the expansion of the action around the “classical” configurations $A_\nu = A_\nu^{\text{cl}}, \phi = \phi^{\text{cl}}$. We shall be interested in the case

$$\phi^{\text{cl}} = 0, \quad (2.4)$$

$$A_\nu^{\text{cl}} = A_\nu^{\text{cl}}(x, y). \quad (2.5)$$

That is, we shall study the effect of external nonuniform fields in a quantum gauge theory. The one-loop contribution to the effective action reads in this case

$$\ln \det M = \ln \det [(-D_\nu^{\text{cl}2} + m^2)/(-\partial_\nu^2 + m^2)], \quad (2.6)$$

with $D_\nu^{\text{cl}} = \partial_\nu - ieA_\nu^{\text{cl}}$.

We shall consider y -independent gauge fields. In the Coulomb gauge they take the form

$$\begin{aligned} A_x &= 0, \\ A_y &= V(x). \end{aligned} \quad (2.7)$$

The effective action can then be written as

$$\Gamma[A_\nu^{\text{cl}}(\cdot)] = L \left[\frac{3}{4} \int \left[\frac{dV}{dx} \right]^2 dx + \frac{1}{2\pi} \int dK \ln \det_x \left[\frac{-\frac{d^2}{dx^2} + m^2 + [K + eV(x)]^2}{-\frac{d^2}{dx^2} + m^2 + K^2} \right] \right], \quad (2.8)$$

where, as usual, we consider the system in a large box of length L . We now introduce dimensionless variables

$$z = ex, \quad \lambda = K/e, \quad \tilde{m} = m/e, \quad (2.9)$$

and then the effective action becomes

$$\begin{aligned} \Gamma[A_\nu^{\text{cl}}(\cdot)] = eL \left[\frac{3}{4} \int \left[\frac{dV}{dz} \right]^2 dz \right. \\ \left. + \frac{1}{2\pi} \int d\lambda \ln \Delta(\lambda) \right], \end{aligned} \quad (2.10)$$

where $\Delta(\lambda)$ is the Fredholm determinant associated with the one-dimensional Klein-Gordon-type equation

tion

$$\left[-\frac{d^2}{dz^2} + \tilde{m}^2 + [\lambda + V(z)]^2 \right] \psi = 0. \quad (2.11)$$

As we stated in the Introduction, we shall obtain a closed solution for a soliton-type field which we choose as

$$V(z) = -\frac{E}{a} (1 - \tanh az), \quad (2.12)$$

where E and a are dimensionless arbitrary parameters. The (dimensionless) electric field is given by

$$E(z) = \frac{E}{\cosh^2 az} \quad (2.13)$$

[The dimensional field is just $eE(z)$.] The field strength is then controlled by E , while $1/a$ determines the spatial extension of the potential. In particular the $a=0$ case corresponds to a uniform electric field of strength eE .

Equation (2.11) has two linear independent solutions, $\psi_1(z)$, and $\psi_2(z)$ for any regular $V(z)$ satisfying the boundary conditions

$$\lim_{z \rightarrow -\infty} V(z) = c > 0, \quad (2.14)$$

$$\lim_{z \rightarrow \infty} V(z) = 0 \quad (2.15)$$

[$c = -2E/a$ for $V(z)$ of the form (2.12)].

For $\text{Re}\lambda > 0$, these solutions have the following asymptotic form:

$$\begin{aligned} \psi_1(\lambda, z) &= e^{-(\lambda+c)z} + S_{12}(\lambda)e^{(\lambda+c)z}, \quad z \rightarrow -\infty \\ \psi_1(\lambda, z) &= S_{11}(\lambda)e^{-\lambda z}, \quad z \rightarrow +\infty \end{aligned} \quad (2.16)$$

$$\begin{aligned} \psi_2(\lambda, z) &= S_{22}(\lambda)e^{(\lambda+c)z}, \quad z \rightarrow -\infty \\ \psi_2(\lambda, z) &= S_{21}(\lambda)e^{-\lambda z} + e^{\lambda z}, \quad z \rightarrow \infty. \end{aligned} \quad (2.17)$$

For the sake of simplicity we have set $m=0$ since this choice does not modify the principal features of the theory.⁸ Analogous expressions hold for $-c \leq \text{Re}\lambda < 0$ and $\text{Re}\lambda < -c$. We shall also need a solution $\tilde{\psi}_1$ in this last region, satisfying the boundary condition

$$\begin{aligned} \tilde{\psi}_1(\lambda, z) &= \tilde{S}_{22}e^{-(\lambda+c)z}, \quad z \rightarrow -\infty \\ \tilde{\psi}_1(\lambda, z) &= e^{\lambda z} + \tilde{S}_{21}e^{-\lambda z}, \quad z \rightarrow \infty. \end{aligned} \quad \text{Re}\lambda < -c \quad (2.18)$$

The Fredholm determinant $\Delta(\lambda)$ can be expressed in terms of the reflection and transmission coefficients $S_{ij}, \tilde{S}_{ij}$:

$$\Delta(\lambda) = [S_{11}(\lambda)S_{22}(\lambda)]^{-1/2}, \quad \text{Re}\lambda > 0 \quad (2.19a)$$

$$\begin{aligned} \Delta(\lambda) &= iS_{21}(\lambda)[(1+c/\lambda)S_{22}(\lambda)]^{1/2}, \\ &\quad -c \leq \text{Re}\lambda < 0 \end{aligned} \quad (2.19b)$$

$$\begin{aligned} \Delta(\lambda) &= \tilde{S}_{21}(\lambda)[(1+c/\lambda)\tilde{S}_{22}(\lambda)]^{-1/2}, \\ &\quad \text{Re}\lambda < -c. \end{aligned} \quad (2.19c)$$

Equations (2.19) can be proved as follows. One starts from the identity

$$\begin{aligned} \frac{d}{d\lambda} [\ln \Delta(\lambda)] &= 2 \int_{-\infty}^{\infty} dz G(z, z; \lambda) \\ &\quad \times [\lambda + V(z)], \end{aligned} \quad (2.20)$$

where $G(z, z'; \lambda)$ is the Green's function for Eq. (2.12), which, for $\text{Re}\lambda \geq 0$, can be written

$$G(z, z'; \lambda) = \frac{\psi_1(z_>, \lambda)\psi_2(z_<, \lambda)}{W[\psi_1, \psi_2]}. \quad (2.21)$$

Here the Wronskian W is defined as $W[f, g] = fg' - f'g$. Expressions analogous to (2.21) hold for negative λ .

Now, if $f(\lambda, z)$ and $g(\mu, z)$ are two solutions of Eq. (2.11) with spectral parameters λ, μ , respectively, then

$$\begin{aligned} \frac{dW}{dz} [f(\lambda, z), g(\mu, z)] \\ = 2f(\lambda, z)g(\mu, z) \left[\frac{\mu + \lambda}{2} + V(z) \right]. \end{aligned} \quad (2.22)$$

With the aid of this last identity the integral in Eq. (2.20) can be evaluated in terms of the asymptotic values of ψ_1, ψ_2 and in this way Eq. (2.19a) follows. An analogous derivation leads to Eq. (2.19b), and (2.19c).

For the solitonlike potential (2.12) the solution of Eq. (2.11) can be expressed in terms of hypergeometric functions, while the coefficients $S_{ij}, \tilde{S}_{ij}$ are ratios of Γ functions. Namely, for $\text{Re}\lambda > 0$,

$$\psi_1(\lambda, z) = S_{11}(\lambda)e^{-Ez/a}(2 \cosh az)^{E/a^2 - \lambda/a} F \left[\frac{\lambda}{a} + p^+, \frac{\lambda}{a} + p^-, \frac{\lambda}{a} + 1; \frac{1}{1 + e^{2za}} \right], \quad (2.23)$$

where

$$p^\pm = \frac{1}{2} - E/a^2 \pm (\frac{1}{4} + E^2/a^4)^{1/2} \quad (2.24)$$

and

$$S_{11}(\lambda) = \frac{\Gamma(\lambda/a + p^+)\Gamma(\lambda/a + p^-)}{\Gamma(\lambda/a + 1)\Gamma(\lambda/a - 2E/a)}, \quad (2.25)$$

$$\begin{aligned} \psi_2(\lambda, z) &= e^{\lambda z}(1 + e^{-2\lambda z})^{E/a^2} \\ &\quad \times F \left[p^+, p^-, 1 - \frac{\lambda}{a}; \frac{1}{1 + e^{2za}} \right], \end{aligned} \quad (2.26)$$

$$S_{21} = - \frac{\Gamma(1-\lambda/a)\Gamma(\lambda/a+p^+)\Gamma(\lambda/a+p^-)}{\Gamma(1+\lambda/a)\Gamma(p^+)\Gamma(p^-)} . \quad (2.27)$$

By computing the appropriate Wronskians, one can obtain relations between the reflection and transmission coefficients. In particular, the following identity holds:

$$S_{11}(\lambda) = (1+c/\lambda)S_{22}(\lambda) . \quad (2.28)$$

III. PROPERTIES OF THE VACUUM IN THE PRESENCE OF A NONUNIFORM ELECTRIC FIELD

We shall compute in this section the effective action in the presence of the nonuniform electric field $E(z)$ given by Eq. (2.13). We shall then introduce the vacuum permeability and study its behavior under different fields.

The one-loop contribution to the effective action

for the nonuniform field $E(z)$ is obtained by performing the integration over λ in Eq. (2.10). Using the explicit expressions for $\Delta(\lambda)$ given by Eqs. (2.19), (2.25), and (2.27), one gets, up to the one-loop order,

$$\Gamma[E(\cdot)] = (2L/ae)\mathcal{L}_{\text{eff}}[E, e; a] , \quad (3.1)$$

$$\mathcal{L}_{\text{eff}} = \frac{e^2 E^2}{2} \left[1 + \frac{e^2}{2\pi} \left[\frac{\partial}{\partial E} \right]^2 G(-E/a^2) \right] . \quad (3.2)$$

We have defined the effective Lagrangian $\mathcal{L}_{\text{eff}}$, so that for a uniform field eE one gets, at the tree level, the usual result

$$\mathcal{L}_{\text{eff}}^{(0)} = \frac{e^2 E^2}{2} .$$

The function $G(K)$ is given by the following expression:

$$\begin{aligned} G(K) = & -2K[\ln\Gamma(K + \frac{1}{2} + \Delta)\Gamma(K + \frac{1}{2} - \Delta) - (K+1-\Delta)\ln(K + \frac{1}{2} - \Delta) \\ & - (K+\Delta)\ln(K + \frac{1}{2} + \Delta) + 2K - \ln 2\pi] + 1 \\ & + 2 \int_0^\infty dt \left[\frac{1}{2} \frac{-1}{t} + \frac{1}{e^+ - 1} \right] (2e^{-t} - e^{-(K+1/2+\Delta)t} - e^{-(K+1/2-\Delta)t}) , \end{aligned} \quad (3.3)$$

$$\Delta = (K^2 + \frac{1}{4})^{1/2} .$$

Equation (2.6) normalizes the effective Lagrangian to zero at zero field. We have still the freedom of a wave-function (finite) renormalization. By introducing a normalization point M , we make the choice

$$\mathcal{L}_{\text{eff}}|_{eE=M} = M^2/2 . \quad (3.4)$$

Under a change of normalization, $E \rightarrow Z^{1/2}(E, e, M)E$, $e \rightarrow Z^{-1/2}(E, e, M)e$ and hence eE is renormalization-group invariant. The wave-function renormalization constant Z has the form

$$Z = 1 + O(e^2/M^2) . \quad (3.5)$$

and the renormalized effective Lagrangian can be written as

$$\mathcal{L}_{\text{eff}} = \frac{e^2 E^2}{2} \epsilon(E, e, M; a) , \quad (3.6)$$

where the vacuum permeability ϵ is given by

$$\begin{aligned} \epsilon = & \epsilon(E, \hat{e} = e/M; a) \\ = & 1 + \frac{1}{2\pi} \hat{e}^2 \left[\left[\frac{\partial}{\partial E} \right]^2 G(-E/a^2) - (1 - \ln 2) \right] , \end{aligned} \quad (3.7)$$

and $\hat{e}$ is a dimensionless coupling constant

$$\hat{e} = \frac{e}{M} , \quad (3.8)$$

from which the appropriate β function is to be defined,

$$\beta \equiv M^2 \frac{\partial \hat{e}}{\partial M^2} . \quad (3.9)$$

Up to one loop it reads⁸

$$\beta = -\hat{e} + \frac{\hat{e}^3}{4\pi} \ln 2 + O(\hat{e}^5) . \quad (3.10)$$

The first negative term is characteristic of two-

dimensional gauge theories, where e has dimensions of a mass. Due to its presence the theory has a stable ultraviolet fixed point at $\hat{e}=0$ and, as we shall see below, this corresponds, exactly as it happens for QCD_4 ,¹⁻³ to an effective action which is controlled by perturbation theory for *large fields*, while for small fields one enters in the strong-

coupling regime and hence perturbative methods cannot be trusted in this region.

In order to discuss the results in terms of the renormalization group, one applies to usual arguments which start from the observation that, under a change of the subtraction point $M_1^2 \rightarrow M_2^2$, the effective action satisfies the functional equation

$$\Gamma[A_\mu, M_1^2, \hat{e}_1] = \Gamma[Z^{1/2}(M_2^2, M_1^2, \hat{e}_1) A_\mu, M_2^2, \hat{e}_2(M_2^2, M_1^2, \hat{e}_1)] . \quad (3.11)$$

One can obtain the differential renormalization-group equation for Γ just by differentiating Eq. (3.11). We shall instead write the corresponding equations for the permeability ϵ and for a related quantity $\tilde{\epsilon}$,

$$\tilde{\epsilon} \equiv (2/M^2) \frac{\partial \mathcal{L}_{\text{eff}}}{\partial E^2} , \quad (3.12)$$

which we define in order to compare more directly our results with those which have been discussed in the literature for Yang-Mills theory.¹⁻³

Both functions $\epsilon(e, E, M; a)$ and $\tilde{\epsilon}(e, E, M, a)$ are dimensionless and they can depend only on dimensionless variables of which two can be found. One is $\hat{e}$, defined by Eq. (3.8) and we choose the other to be

$$t = \ln \hat{e} E . \quad (3.13)$$

We are considering the spatial extension of the potential ($1/a$) as a parameter which will play no role in the renormalization-group analysis that follows. In terms of these variables the renormalization-group equations take the form

$$-\frac{\partial \epsilon}{\partial t}(\hat{e}(t), t) + \beta(\hat{e}) \frac{\partial \epsilon}{\partial \hat{e}} = 0 , \quad (3.14)$$

$$-\frac{\partial \tilde{\epsilon}}{\partial t}(\hat{e}(t), t) + \beta(\hat{e}) \frac{\partial \tilde{\epsilon}}{\partial \hat{e}} - [2\gamma(\hat{e}) - 1] = 0 , \quad (3.15)$$

where the effective coupling constant satisfies

$$\frac{d\hat{e}(t)}{dt} = \beta(\hat{e}(t)) \quad (3.16)$$

with the boundary condition

$$\hat{e}(0) = \hat{e} \quad (3.17)$$

and the anomalous dimension γ is given by

$$\gamma = -M^2 \frac{\partial \ln Z}{\partial M^2} = \frac{\beta(\hat{e})}{\hat{e}} + \frac{1}{2} . \quad (3.18)$$

Equation (3.14) implies that ϵ is an invariant

charge,

$$\epsilon(\hat{e}(t), t; a) = \epsilon(\hat{e}, 0; a) . \quad (3.19)$$

Concerning Eq. (3.15), its solution takes the form

$$\tilde{\epsilon}(\hat{e}(t), t; a) = \frac{\hat{e}^2}{\hat{e}^2(t)} \tilde{\epsilon}(e, 0, a) . \quad (3.20)$$

This expression, valid for a two-dimensional gauge theory, coincides with the remarkable formulas first given by Mantinyan and Savvidy¹ for QCD_4 and QED_4 . It establishes the relation between the asymptotic behavior of the effective Lagrangian for an intense field and that corresponding to small distances, controlled by the β function. Hence, also in this model β decides the density of the vacuum energy in the presence of a background field.

Our results for nonuniform fields can now be compared with those already obtained for uniform field.⁸ (Notice the difference with Ref. 8 in the definitions of the permeabilities ϵ and $\tilde{\epsilon}$, due to the fact that in this investigation we are using a dimensionless field strength E .)

We have calculated from Eqs. (3.7) and (3.3) the values of ϵ as a function of E for different values of $\hat{e}$ and a . In Fig. 1, we have plotted the results for fixed $\hat{e} = \frac{1}{2}$ and different values of a . As we already stated, the limiting case $a=0$ corresponds to uniform field and of course we reproduce the results already obtained in Ref. 8: there exists a critical field $E_c(a=0)$ such that for $E > E_c$, $0 \leq \epsilon < 1$. Because of our normalization condition (3.4), the critical field corresponds to the value

$$E_c(a=0) = 1/\hat{e} . \quad (3.21)$$

Now, for nonuniform fields of the form (2.13), we find that the vacuum permeability is smaller than one for electric field strengths above a certain critical value. That is, the nontrivial structure of the vacuum is not restricted to uniform background fields. The precise value of the critical strength depends now on the spatial extension of

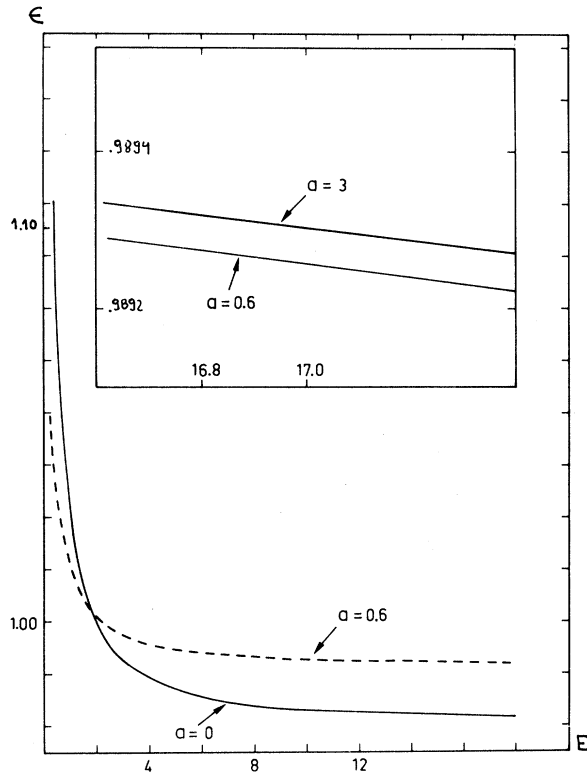


FIG. 1. The permeability as a function of the field strength for fixed charge ($\hat{e} = \frac{1}{2}$). The solid curve on the bottom corresponds to uniform field ($a=0$) and the dashed one to $a=0.6$. In the small rectangle at the top an enlarged scale is used to show how the results change with the spatial extension.

the background field and becomes larger for large values of a (smaller extension). That is, the less extended is the field, the stronger has to be its strength in order to turn the vacuum into a dielectric medium. An associated effect is also clear from Fig. 1: The antiscreening decreases for growing a , although one has to take large values of a in order to see a significant effect. This can be understood, noting that the one-loop contribution to

ϵ , Eq. (3.7), depends on E and a through the ratios E/a and E/a^2 and hence, since for increasing E , ϵ decreases, the same effect will occur for decreasing a .

We conclude our discussion by noting that the use of spectral methods allowed us to compute the effective action in a closed form up to one-loop order for a two-parameter dependent nonuniform potential. This shows the possibility of developing a scheme based on the Gelfand-Levitan equation in order to consider more general one-dimensional fields depending on an arbitrary number of parameters. Concerning the particular model that we analyzed, our computations show that the non-trivial structure of the vacuum—its dielectricity, which leads naturally to confinement^{2,9}—is not due to the uniformity of the used background field, but on the contrary it is inherent to scalar electrodynamics in two space-time dimensions, a model which shares many relevant features with more realistic non-Abelian theories in four dimensions.

Concerning the last theories, it is worthwhile to notice that the spectral method are not limited to one-dimensional problems. The inverse-scattering transformation can be used to deal with rotationally invariant problems in an arbitrary number of dimensions⁶ ν and also to study one-variable dependent fields immersed in a ν -dimensional space-time. For example, the extension of our analysis to scalar electrodynamics in $\nu=4$ dimensions can be done by repeating the steps outlined in Sec. II, except that the Klein-Gordon-type Eq. (2.11) reads in this case

$$\Delta(K_{\perp}, \lambda, A)\psi = \left[-\frac{d^2}{dx^2} + m^2 + K_{\perp}^2 + [\lambda + A_y(x)]^2 \right] \psi = 0, \quad (3.22)$$

where $K_{\perp}$ is the momentum in the directions other than x and y .

Then, the one-loop contribution to the effective action is given by

$$\Gamma^{(1)} = 4 \frac{L^{\nu-1}}{\Gamma\left[\frac{\nu-2}{2}\right] (4\pi)^{\nu/2}} \int_0^{\infty} K_{\perp}^{\nu-3} dK_{\perp} \int_0^{\infty} d\lambda \ln \det \frac{\Delta(K_{\perp}, \lambda, A)}{\Delta(K_{\perp}, \lambda, 0)}. \quad (3.23)$$

For potentials of the form (2.13), $\det_x$ has a closed (although slightly more complicated) expression as that given by Eq. (3.3). This and other related problems are currently under study.

ACKNOWLEDGMENTS

F. A. Schaposnik wishes to acknowledge the members of the Laboratoire de Physique Théorique et Hautes Energies de l'Université Pierre et Marie Curie for their hospitality.

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